3rd-Party Libraries, Assets, Licensing & Implementation Status

1. 3rd-Party Libraries and Build Tools (UR_LIBRARY, QR_LICENCE)

Library / Tool	Purpose in System	Licence	Suitability & Sustainability Analysis
libGDX (gdx, lwjgl3, box2d, freetype, tools)	Core game engine, 2D physics, and font rendering	Apache Licence 2.0	Fully Suitable: Permissive licence granting modification, distribution, and patent rights without copyleft obligations.
Ashley, gdx-ai & Box2DLights	ECS architecture, AI utilities, and 2D lighting	Apache Licence 2.0	Fully Suitable: Compatible with libGDX; permits closed-source or commercial resale by the customer.
LWJGL 3 (glfw, opengl, openal, stb, jemalloc)	Low-level native bindings for graphics and audio	BSD 3-Clause	Fully Suitable: Academic permissive licence allowing commercial redistribution if copyright notices are retained.
Construo (io.github.fourlastor)	Build plugin for packaging native executables	Apache Licence 2.0	Fully Suitable: Build-time tool only; imposes no licensing conditions on distributed game binaries.

2. 3rd-Party Assets (QR_LICENCE)

Asset File	Source	Licence / Terms	Suitability & Sustainability Analysis
PixelifySans VariableFont_wght.ttf	Google Fonts	SIL Open Font Licence (OFL) 1.1	Suitable: Open-source font licence permitting bundling, modification, and commercial distribution.
player.png	Avatars In Pixels (Minipix Maker)	Proprietary (Attribution; Non Commercial)	Prototype Only (Unsustainable): Acceptable for Assessment 1 academic evaluation, but violates commercial resale rules (QR_LICENCE). To be replaced with a CC0/custom sprite in Assessment 2.
bgd.png / bgd2.jpg (Grass textures)	iStock (Credit: Nastco)	Proprietary Stock (Paid Licence required)	Prototype Only (Unsustainable): Watermarked previews used temporarily. Will be replaced with CC0 or in-house textures in Assessment 2 to avoid paid dependencies (QR_LICENCE).

3. Licence Compatibility & Overall Sustainability

In accordance with UR_LIBRARY and QR_LICENCE, all software dependencies use mutually compatible permissive licences (Apache 2.0 and BSD 3-Clause) free of copyleft restrictions (unlike GPL). This supports long-term maintainability (UR_MAINTAIN) for the team inheriting the codebase in Assessment 2 and enables future commercial distribution by the customer. Licensing risks are strictly isolated to the temporary third-party image assets (player.png, bgd.png, bgd2.jpg) decoupled inside assets/ and scheduled for replacement.

4. Unimplemented / Partially Implemented Assessment 1 Features

UR_INV / FR_INVINT, FR_ITEMUSE, FR_ITEMEFFECT, FR_ITEMCOUNT, FR_ITEMVALID (Inventory & Item Usage): The initial two inventory items and their dispersal effects on geese are not yet implemented in the Assessment 1 build.

UR_G_MOVE / FR_GEESE, FR_MOVESPEED (Geese Entities & Movement): Goose sprites (assets/Goose) exist in the repository, but active goose entities and chasing behaviour are not yet integrated into GameScreen. UR_COUNTER / FR_COUNTERS (Usage & Event Counter): The on-screen counter tracking items used and hidden events triggered is partially implemented and pending UI integration.

UR_WINCOND, UR_RESULT / FR_WIN, FR_TIMEOUT, FR_RESULTSCREEN (Win/Loss & End Screen): While GameTimer tracks the 5-minute countdown (FR_TIMEDISPLAY) and LakeEvent/PitEvent implement hidden events (FR_EVENTHIDDEN), the game-over/victory state transition screen is not yet implemented.